

Klastoets 2

Kopiereg voorbehou

Class Test 2

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Module EBN111

Elektrisiteit en Elektronika

13 April 2012

Module EBN122

Electricity and Electronics

13 April 2012



**UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA**

Toetsinligting:

Test information:

Maksimum punte: <i>Maximum marks:</i>	16	Volpunte: <i>Full marks:</i>	15
Duur van vraestel: <i>Duration of paper:</i>	30 minute <i>30 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>	8		

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen.
Students may do their own rough calculations on the question paper.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert
	2 JD le Roux

Groephef: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding.

Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.

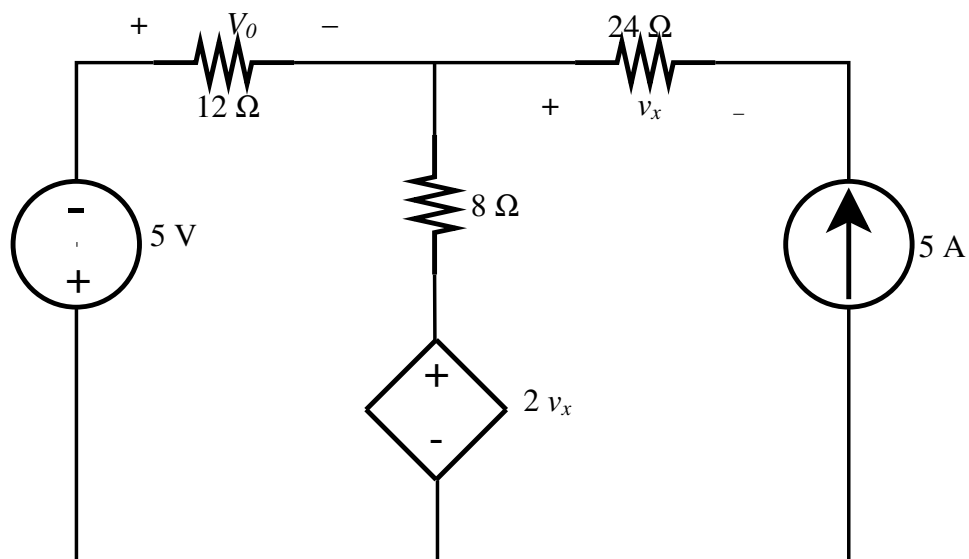
Groephef: <i>Group Head:</i>	Prof J Joubert	Handtekening: <i>Signature:</i>	
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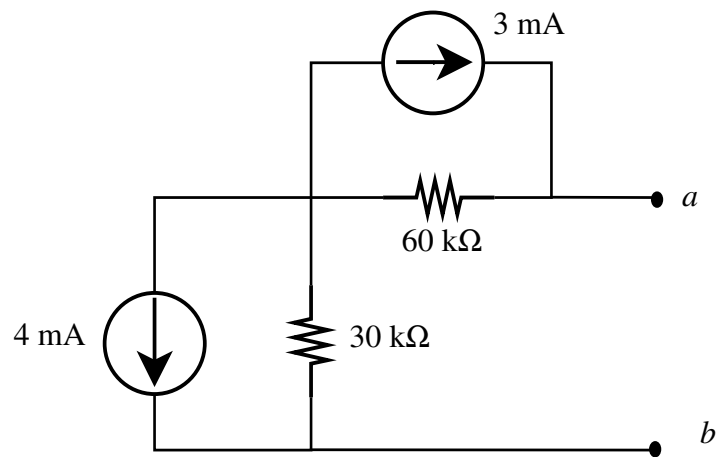
Vraag 1.1 [2]

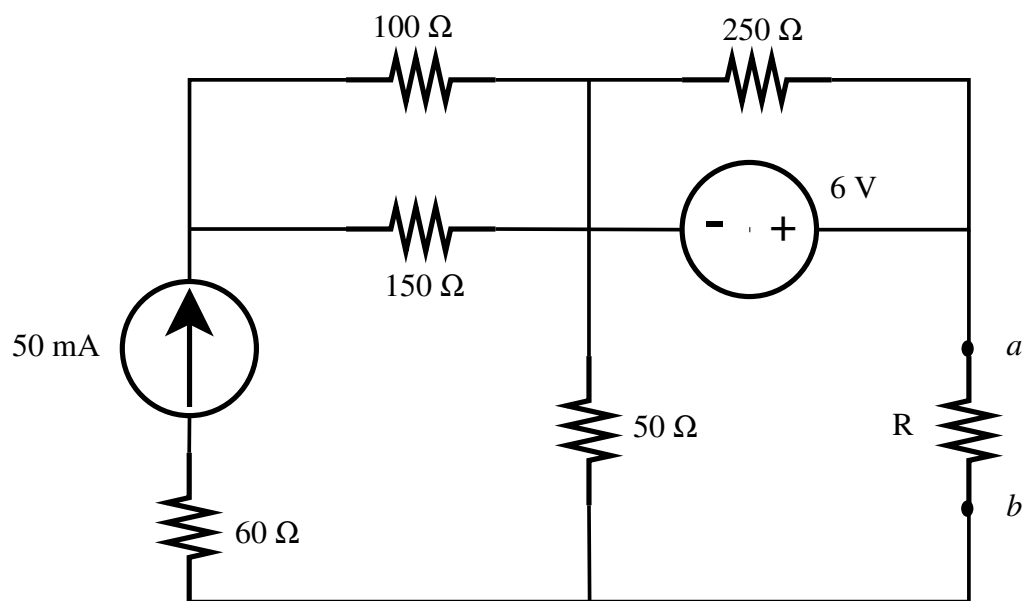
Bepaal die spanning V_o slegs as gevolg van die 5 V spanningsbron.

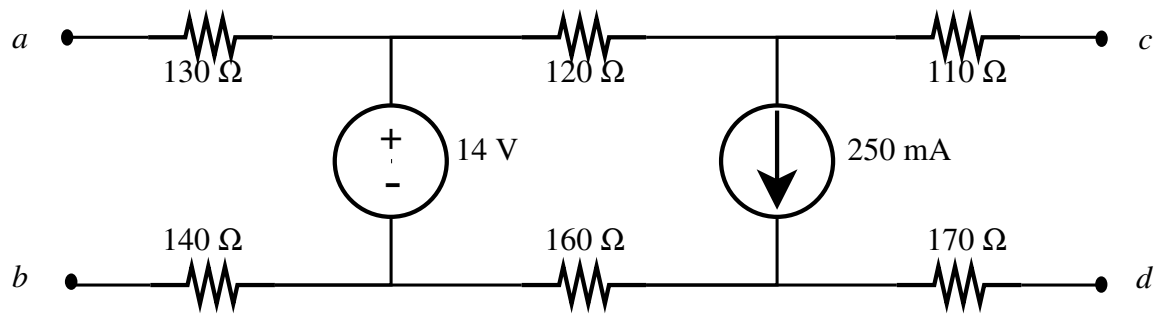
Question 1.1 [2]

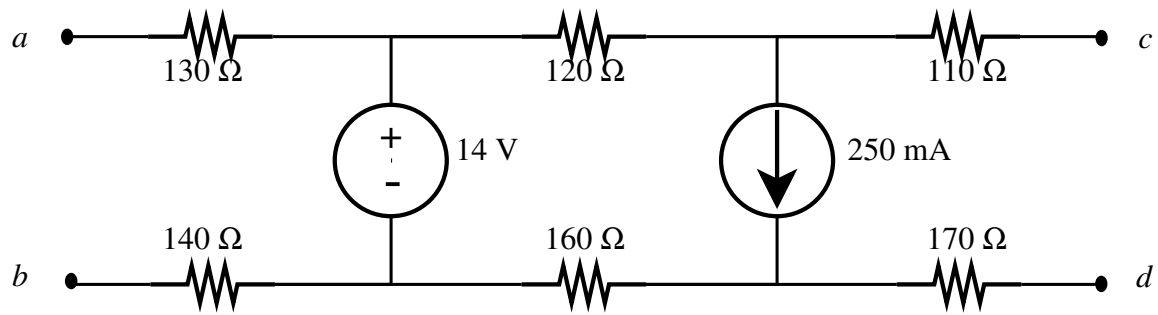
Determine the voltage V_o only due to the 5 V voltage source.



Vraag 1.2 [3]Maak gebruik van brontransformatie om V_{ab} te bereken.**Question 1.2 [3]**Make use of source transformation to determine V_{ab} .

Vraag 1.3 [4]Bepaal V_{TH} en R_{TH} soos gesien deur terminale $a-b$.**Question 1.3 [4]**Determine V_{TH} and R_{TH} as seen by terminals $a-b$.

Vraag 1.4 [2]Bepaal I_N soos gesien deur terminale c - d .**Question 1.4 [2]**Determine I_N as seen by terminals c - d .

Vraag 1.5 [2]Bepaal V_{TH} soos gesien deur terminale a - b .**Question 1.5 [2]**Determine V_{TH} as seen by terminals a - b .

Vraag 1.6 [3]Bepaal R_{TH} soos gesien deur terminale a - b .**Question 1.6 [3]**Determine R_{TH} as seen by terminals a - b .